SHARP

Worksheet 2 – Algebra: Multiplying and Factorising Expressions

Grade 10 Mathematics

1. Identify whether the following expressions are monomial, binomial, trinomial or polynomial:

a) $2x^2 + 3$ b) $2x^2 \times 3$

c) $2x^2 + 3x + 1$ d) $2x^2 + 3x^3 - x + 4$

e) $x^2 - 5x + 6$ f) $4(3x^2 + 3)$

g) $7x$ h) $\frac{7x^2+4}{x}$

i) $x^3 - 2x^2 + 3x - 4$ j) $2x^2 + 11x - 12$

2. Multiply out and then simplify the following:

a) $(x + y)(x - y)$ b) $(x + y)^2$

c) $(x + y)(x^2 + 2xy + y^2)$ d) $\left(\frac{1}{2}x + \frac{1}{3}y\right)(2x + 3y)$

e) $(x + y)(x^2 - xy + y^2)$ f) $(2a + b)(2b + a)$

g) $(4a - 5b)(4a + 5b)$ h) $(a + 2b)(3a^2 - 4ab + 5b^2)$

i) $(4a - 5b)(16a + 20ab + 25b^2)$ j) $(3a + 4b)^2$

k) $\left(\frac{2}{3}x - \frac{3}{4}y\right)\left(\frac{3}{2}x + \frac{4}{3}y\right)$ l) $(6c + 9d)(6c - 9d)$

m) $(3x + 5y)(3x + 5y)$ n) $(2c - 5d)(3cd + 4d^2 - 6c^2)$

o) $\left(\frac{1}{4}g + \frac{2}{3}\right)(16g - 9)$ p) $(2c - 5d)(4c^2 + 10cd + 25d^2)$

q) $(3h - 5d)^2$ r) $(2x - 3y)(2x + 3y)$

s) $(7d + 3e)(49d^2 - 21de + 9e^2)$ t) $3\left(\frac{5}{6}x - \frac{1}{2}y\right)\left(3x + \frac{1}{4}y\right)$

3. Factorise the following by finding the common factor:

a) $3x^2 + 6y^2$ b) $3x^3 + 6xy^2$

c) $3a^2b - 4ab^2 + 12ab$ d) $11abc^2 - 22a^2b + 33ab^2d$

e) $\frac{3}{2}x^2 + \frac{5}{2}xy - \frac{1}{2}y^2$ f) $12a^3b^2c - 16a^2b^2 + 24a^4b^5c^3$

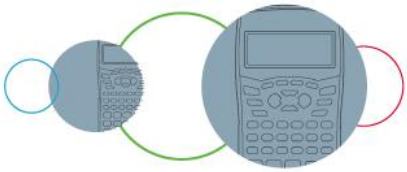
g) $13a^4b - 26ba^3 + 39a^2b^2$ h) $35a + 7ab - 21a^2b + 14abc$

i) $4abc + 16ab - 20bc + 24ac$ j) $3a(a + b) - 4b(a + b)$

4. Factorise the following expressions:

a) $x^2 - y^2$ b) $36x^2 - 49$

c) $81a^2 - 16b^2$ d) $a^4 - b^4$



e) $64a^4 - b^4$ f) $9 + 36c^2$

g) $(a+b)^2 - c^2$ h) $50x^3 - 18xy^2$

i) $12a^3b - 300ab^3$ j) $d^2 - (3a+b)^2$

5. Factorise the following perfect square trinomials:

a) $a^2 + 4a + 4$ b) $9a^2 - 30ab + 25b^2$

c) $x^2 - 24x + 144$ d) $d^2 + d + \frac{1}{4}$

e) $x^2 + 10x + 25$ f) $x^2 - 12x + 36$

g) $x^2 - 18xy + 81y^2$ h) $x^2 + 16xy + 64y^2$

i) $9a^2 + 12ab + 4b^2$ j) $25c^2 - 60cd + 36d^2$

6. Factorise the following trinomials:

a) $a^2 + a - 20$ b) $x^2 + 6x + 8$

c) $y^2 - 9y + 8$ d) $b^2 + 8ab + 15a^2$

e) $a^2 + 9ab - 22b^2$ f) $x^2 + 2xy - 3y^2$

g) $x^2 + 1\frac{1}{2}x - 1$ h) $y^2 + 11y + 24$

i) $x^2 - 14x + 48$ j) $a^2 + 2\frac{2}{3}a + \frac{4}{3}$

7. Factorise the following by grouping:

a) $ac + bd + ad + bc$ b) $2a^3 + 6ab + a^2b + 3b^2$

c) $a^3c - 12b^2d - 4abc + 3a^2bd$ d) $a^2c^2 - 2bc^2 + 3a^2d - 6bd$

e) $30b^2d^4 - 6b^2c^3 - 15ad^4 + 3ac^3$ f) $2bde + 6ab - 14cde - 42ac$

g) $2x^3y - 6xy^3 - 3y^4 + x^2y^2$ h) $28xy^2 - 21y^3 + 8x^3 - 6x^2y$

i) $5c^3d^2hg^2 - 5c^4d^2e - 5cdefg^2h + 5c^2de^2f$

j) $7a^3c^3 + 7b^2c^3 + 7a^3cde + 7b^2cde$

8. Factorise the following cubes:

a) $x^3 + y^3$ b) $x^3 - y^3$ c) $8a^3 - 27y^3$

d) $64b^6 + 125$ e) $2ab^4 - 16a^4b$ f) $81x^3 + 24y^3$

g) $(a+b)^3 + c^3$ h) $c^3 - (a+2b)^3$ i) $250 - 54x^3$

